

Muhang Tian

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EDUCATION

Courant Institute, New York University
Ph.D. in Computer Science

01/2025 – Expected 12/2029

Duke University
B.S. in Computer Science, double minor in Mathematics and Economics
Graduation with Highest Distinction, Alex Vasilos Memorial Award

08/2020 – 05/2024

PUBLICATIONS

- [1]

Multi-objective Reinforcement Learning with Nonlinear Preferences: Provable Approximation for Maximizing Expected Scalarized Return
Nianli Peng*, [Muhang Tian*](#), Brandon Fain
Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)

2025
[\[Link to paper\]](#)
- [2]

Reliable Generation of Privacy-preserving Synthetic Electronic Health Record Time Series via Diffusion Models
[Muhang Tian](#), Bernie Chen, Allan Guo, Shiyi Jiang, Anru R Zhang
Journal of the American Medical Informatics Association (JAMIA)

2024
[\[Link to paper\]](#)
- [3]

Fast and Interpretable Mortality Risk Scores for Critical Care Patients
Chloe Qinyu Zhu*, [Muhang Tian*](#), Lesia Semenova, Jiachang Liu, Jack Xu, Joseph Scarpa, Cynthia Rudin
Journal of the American Medical Informatics Association (JAMIA)

2024
[\[Link to paper\]](#)
- [4]

Welfare and Fairness in Multi-objective Reinforcement Learning
Zimeng Fan*, Nianli Peng*, [Muhang Tian*](#), Brandon Fain
Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)

2023
[\[Link to paper\]](#)

WORK EXPERIENCE

- ByteDance**
Research Intern

06/2026 – 09/2026

Explored an agentic framework to automate research experiments for generative recommender systems.
- Big Ideas Lab**
Machine Learning Engineer

05/2022 – 03/2023

Applied EfficientDet for object detection tasks on COVID-19 test results.
- Afloat**
Software Engineer

05/2022 – 08/2022

Maintained company server, performed software testing, and reported weekly directly to co-founders.
- Sinolink Securities**
Equity Research Analyst

05/2021 – 08/2021

Analyzed market data for target firms, wrote weekly reports, and managed the company's database.

RESEARCH EXPERIENCE

- K-space Transformer**

03/2025 – 03/2026

- Explored using Vision Transformers to learn meaningful representations from k-space, aiming to perform prediction directly from Fourier transforms of natural images.
 - Explored on ImageNet with different training recipes and normalization schemes. Observed that using a linear layer for tokenization is crucial for accurate prediction.

Diffusion Model for Synthetic Health Records

05/2023 – 12/2023

- Explored generating mixed-type time series data with multinomial and Gaussian diffusions using bidirectional recurrent neural network as the backbone.
- Experimentally demonstrated the proposed approach produces synthetic samples with high data utility, outperforms generative adversarial networks, and improves ML models' performance.

Nonlinear Multi-objective Reinforcement Learning

05/2022 – 05/2024

- Explored optimizing the expectation over a nonlinear objective on the cumulative reward. Proposed an approximate algorithm that optimizes this objective by considering past accumulated rewards.
- Experimentally and theoretically verified that the algorithm converges. Designed another dynamic programming approach to learn a policy with theoretical guarantees.

Interpretable Machine Learning for Risk Prediction

01/2023 – 10/2023

- Explored finding an interpretable model for classification task through approximately optimizing a L0 regularized logistic regression model with integer, sparsity, and box constraints.
- Designed group sparsity and monotonicity constraints to improve readability and usefulness of the risk score in practical settings.

Bandit and Reinforcement Learning Algorithms

10/2022 – 03/2023

- Researched Sample Average Uncertainty approach towards multi-arm bandit problems by utilizing approximate sample mean variance as an exploration bonus.
- Implemented Deep Q-Learning, Proximal Policy Optimization (PPO), and Advantage Actor-Critic algorithms from scratch and compared their performances in Atari environments.

HONORS AND AWARDS

Henry M. MacCracken Award: multi-year funding for New York University PhD students. 2024
CS+ Summer Research Fellowship: Duke summer research funding (\$10,000). 2022 – 2023

ACADEMIC SERVICES

Reviewer: NeurIPS (2023), ICLR (2024), AAAI (2025), IEEE ICHI (2023).
Teaching Assistant: Fundamentals of Machine Learning, Design and Analysis of Algorithms, Econometrics.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, R, SQL, MATLAB, CSS/HTML.
Softwares: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Linux, Slurm, Docker, Git, Linux.